
UNIT 6 DESCRIPTIVE AND EXPLORATORY DATA ANALYSIS – AN INTRODUCTION

Structure

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6.0 INTRODUCTION

Building on the foundational understanding of data science principles (Unit 1), data types and structures (Unit 2), data acquisition methods (Units 3 and 4), and data preprocessing techniques (Unit 5), this Unit i.e. Unit 6 marks a significant transition into the analytical phase of the data science workflow. Once data has been collected and cleaned, the next logical step is to begin exploring and summarizing it to uncover patterns, trends, and initial insights.

Unit 6, “Descriptive and Exploratory Data Analysis – An Introduction,” provides the learner with tools and techniques to perform preliminary analysis that reveals the structure and behavior of the data. This unit emphasizes the role of descriptive statistics—such as measures of central tendency, variability, and distribution—and introduces exploratory methods that help in understanding data relationships without formal modeling. These techniques act as a diagnostic toolkit, allowing analysts to ask meaningful questions and guide the selection of further statistical or predictive methods.

The knowledge gained from Block-1 enables students to recognize the importance of data characteristics and quality, while Unit 5 ensures that the data used in this unit is well-prepared for meaningful analysis. Together, they set the stage for Unit 6, where learners begin to derive initial interpretations, generate hypotheses, and visualize data behaviour—critical capabilities in both academic research and real-world analytics projects.

By the end of this unit, learners will be equipped not just with analytical techniques, but with the ability to think critically about what the data is suggesting. This unit thus acts as a bridge between data preparation and more advanced analytical reasoning, ensuring a smooth and logical progression in the data science learning pathway.

This Unit introduces you to the basic concept of data sciences. This Unit provides an introduction to different types of data used in data science. It also points to other kinds of analysis that can be performed using data science. Further, the Unit also introduces some of the common mistakes of data science.

6.1 OBJECTIVES

At the end of this Unit, you should be able to:

- Define the term data science in the context of an organisation
- Explain different types of data
- Explain Descriptive analysis
- Explain Exploratory analysis

6.2 DATA SCIENCE - DEFINITION

Data Science is a multidisciplinary science with the objective of performing data analysis to generate knowledge that can be used for decision-making. This knowledge can be in the form of similar patterns or predictive planning models, forecasting models etc. A data science application collects data and information from multiple heterogeneous sources, cleans, integrates, processes and analyses this data using various tools and presents information and knowledge in various visual forms.

As stated earlier, data science is multidisciplinary, as shown in Figure 1.

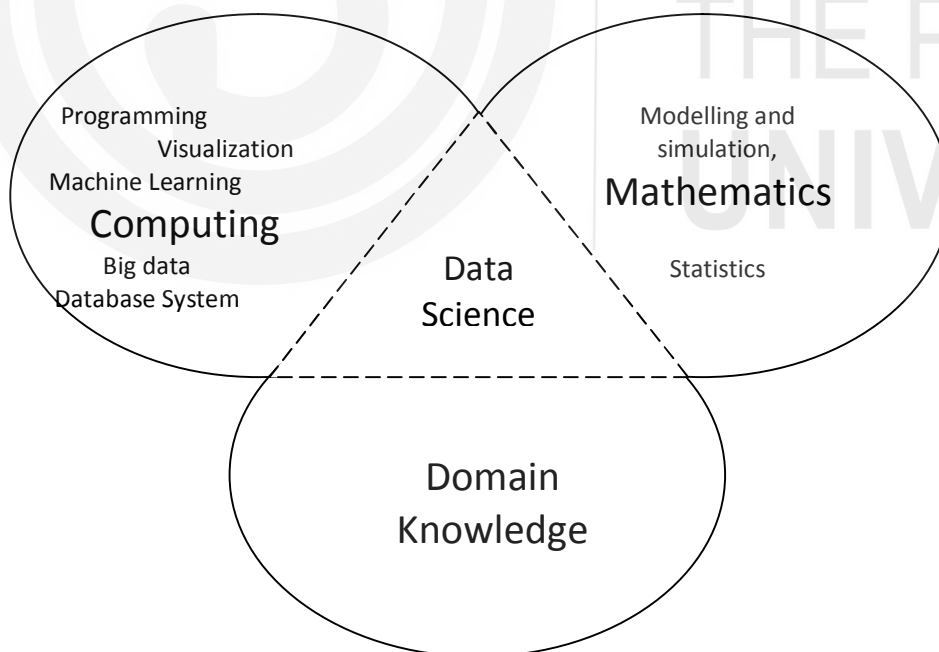


Figure 6.1: Data Science

What are the advantages of Data science in an organisation? The following are some areas in which data science can be useful.

- It helps in making business decisions, such as deciding the health of companies with whom they plan to collaborate,

- It may help make better predictions for the future, such as creating strategic plans for the company based on present trends etc.
- It may identify similarities among various data patterns leading to applications like fraud detection, targeted marketing etc.

Data science is a way forward for business decision-making, especially in the present-day world, where data is generated at the rate of Zetta bytes.

Data Science can be used in many organisations. Some of the possible uses of data science are given below:

- It has excellent potential for finding the best dynamic route from a source to a destination. Such applications may constantly monitor the traffic flow and predict the best route based on collected data.
- It may bring down the logistic costs of an organisation by suggesting the best time and route for transporting foods
- It can minimise marketing expenses by identifying similar group buying patterns and performing selective advertising based on the data obtained.
- It can help in making public health policies, especially in the case of disasters.
- It can be useful in studying the environmental impact of various developmental activities
- It can be very useful in saving resources in smart cities

6.3 TYPES OF DATA

The type of data is one of the essential aspects determining the kind of analysis that can be performed on data. In data science, the following are the different types of data that are required to be processed:

1. Structured Data
2. Semi-Structured Data
3. Unstructured data
4. Data Streams

Structured Data

Since the start of the era of computing, the computer has been used as a data processing device. However, it was not before the 1960s when businesses started using computers to process their data. One of the most popular languages of that era was the **Common Business-Oriented Language (COBOL)**. COBOL had a data division representing the structure of the data being processed. This was followed by a disruptive seminal design of technology by E.F. Codd. This led to the creation of relational database management systems (RDBMS). RDBMS allows structured storage, retrieval and processing of integrated data of an organisation that can be securely shared among several applications. The RDBMS technology also supported secure transactions, thus, became a major source of data generation. Figure 2 shows the sample structure of data that may be stored in a relational database

system. One of the key characteristics of structured data is that it can be associated with a schema. In addition, each schema element may be related to a specific data type.

Customer (custID, custName, custPhone, custAddress, cost Category, custPAN, custAadhar)

Account (AccountNumber, custIDoffirstaccountholder, AccountType, AccountBalance)

JointHolders (AccountNumber, custID)

Transaction(transDate, transType, AccountNumber, Amountoftransaction)

Figure 6.2: A sample schema of structured data

Relational data is structured data. Many organisations are collecting a large amount of this structured data as the backend of their applications. The concept of a data warehouse was introduced in the 90s. A data warehouse is a time-invariant, subject-oriented aggregation of data of an organisation that can be used for decision-making. Data in a data warehouse is represented using dimension tables and fact tables. The dimensional tables classify the data of fact tables. The data of the data warehouse is also structured in nature and can be used for analytical data processing and data mining. In addition, many different types of database management systems have been developed, mostly storing structured data.

However, with the growth of communication and mobile technologies, many different applications became very popular, leading to the generation of a very large amount of semi-structured and unstructured data. These are discussed next.

Semi-structured Data

As the name suggests, Semi-structured has some structure in it. The structure of semi-structured data is due to the use of tags or key/value pairs. The common form of semi-structured data is produced through XML, JSON objects, Server logs, EDI data, etc. An example of semi-structured data is shown in Figure 3.

<pre><Book> <title>Data Science and Big Data</title> <author>R Raman</author> <author>C V Shekhar</author> <yearofpublication>2020</yearofpublication> </Book></pre>	<pre>"Book": { "Title": "Data Science", "Price": 5000, "Year": 2020 }</pre>
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Figure 6.3: Sample semi-structured data

Unstructured Data

The unstructured data does not follow any schema definition. For example, the written text-like content of this Unit is unstructured. You may add specific headings or metadata for unstructured data. In fact, the growth of the Internet has resulted in the generation of Zetta bytes of unstructured data. Some of the unstructured data is listed below:

- A large amount of written textual data such as email data, social media data, etc.
- Unprocessed audio and video data
- Image data and mobile data
- Unprocessed natural speech data
- Unprocessed geographical data

This data generally requires enormous storage space, newer processing methods and faster processing capabilities.

Data Streams

A data stream is characterised by a sequence of data over a period of time. Such data may be structured, semi-structured or unstructured, but it gets generated repeatedly. For example, IoT devices like weather sensors will generate a data stream of pressure, temperature, wind direction, wind speed, humidity etc., for a particular place where it is installed. Such data is huge for many applications that are required to be processed in real-time. In general, not all the data of streams need to be stored, and such data is required to be processed for a specific duration of time.

6.3.1 Statistical Data Types

Two distinct types of data can be used in statistical analysis. These are – Categorical data and Quantitative data.

Categorical or qualitative Data:

Categorical data identifies the category of data. For example, the occupation of a person may take values of the categories "Business", "Salaried", "Others", etc. The categorical data can be of two distinct measurement scales called Nominal and Ordinal, which are given in Figure 4. If the categories are not related, then categorical data is of Nominal data type. For example, the Business category and Salaried categories have no relationship; therefore, it is of Nominal type. However, a categorical variable like age category, defining age in categories "0 or more but less than 26", "26 or more but less than 46", "46 or more but less than 61", and "More than 61" has a specific relationship. For example, a person in the "More than 61" age category is older than anyone in any other age category.

Quantitative Data:

Quantitative data is numeric data that can be used to define different scales of data. Qualitative data is also of two basic types –discrete, representing distinct numbers like 2, 3, 5,...; or continuous, representing continuous values of a given variable. For example, your height can be measured using a continuous scale.

Measurement scale of data

Data are raw facts; for example, student data may include name, Gender, Age, Height of student, etc. The name typically is a distinguishing data that tries to distinctly identify two data items, just like the primary key in a

database. However, the name data or any other identifying data may not be useful for data analysis in data science. The data such as Gender, Age, and Height may be used to answer queries of the kind: Is there a difference in the height of boys and girls in the age range 10-15 years? One of the important questions is how you measure the data so that it is recorded consistently. Stanley Stevens, a psychologist, defined the following four characteristics of any scale that can be measured:

- Every representation of the measure should be unique, referred to as the identity of a value (*IDV*).
- The second characteristic is the magnitude (*M*), which clearly can be used to compare the values; for example, a weight of 70.5 kg is more than 70.2 kg.
- The third characteristic is about equality in intervals (*EI*) used to represent the data; for example, the difference between 25 and 30 is 5 intervals, which is the same as the difference between 41 to 46, which is also 5 intervals.
- The last characteristic is about a defined minimum or zero value (*MZV*); for example, in the Kelvin scale, the temperature has an absolute zero value, whereas the Intelligent quotient cannot be defined as zero.

Based on these characteristics, four basic measurement scales are defined. Figure 4 illustrates these measurements, their features and examples.

Measurement Scale	Characteristics				Example
	<i>IDV</i>	<i>M</i>	<i>EI</i>	<i>MZV</i>	
Nominal	Yes	No	No	No	Gender F - Female M- Male
Ordinal	Yes	For rank ordering	No	No	A hypothetical Income Category: 1 - "0 or more but less than 26" 2 - "26 or more but less than 46" 3 - "46 or more but less than 61" 4 - "More than 61"
Interval	Yes	Yes	Yes	No	IQ, Temperature in Celsius
Ratio	Yes	Yes	Yes	Yes	The temperature in K, Age

Figure 6.4: Measurement Scales of Data

6.3.2 Sampling

In general, the size of the data that is to be processed is quite large. This leads you to the question: whether you would use the entire data or some representative sample of this data?. In several data science techniques, sample data is used to develop an exploratory model. Thus, even in data science, sampling is one of the ways to enhance the speed of exploratory data analysis. In this case, the population may be the entire data set you may be interested in. Figure 5 shows the relationships between the population and the sample. One of the questions that can be asked in this context is, what should

be the size of a good sample? You may have to find the answer in the literature. However, you may please note that a good sample is representative of its population.

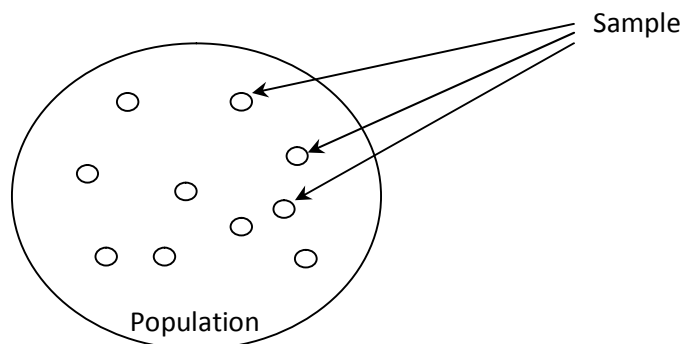


Figure 6.5: Population and Sample

One of the main objectives of statistics, which uses sample data, is to determine the statistics of the sample and find the probability that the statistic developed for the sample would determine the parameters of the population with a specific percentage of accuracy. Please note the terms stated above are very important and explained in the following table:

Term	Used for	Example
Statistic	The statistic is computed for a sample	Sample mean ($\bar{x}$), Sample Standard deviation (s), Sample size (n)
Parameter	Parameters are predicted from a sample and are about the population	Population mean (μ), Population Standard deviation (σ), Population size (N)

Next, we discuss different kinds of analysis that can be performed on data.

Check Your Progress 1:

1. Define the term data science.

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2. Differentiate between structured, semi-structured, unstructured and stream data.

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3. What would be the measurement scale for the following? Give reasons in support of your answer.

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6.4 BASIC METHODS OF DATA ANALYSIS

The data for data science is obtained from several data sources. This data is first cleaned of errors, duplication, aggregated and then presented in a form that can be analysed by various methods. This section defines some of the basic techniques used for analysing data. These are Descriptive analysis, Exploratory data analysis and Inferential data analysis.

6.4.1 Descriptive Analysis

Descriptive analysis is used to present basic data summaries; however, it does not attempt to interpret the data. These summaries may include different statistical values and certain graphs. Different types of data are described using different ways. The following example illustrates this concept:

Example 1: Consider the data given in the following Figure 6. Show the summary of categorical data in this Figure.

Enrolment Number	Gender	Height
S20200001	F	155
S20200002	F	160
S20200003	M	179
S20200004	F	175
S20200005	M	173
S20200006	M	160
S20200007	M	180
S20200008	F	178
S20200009	F	167
S20200010	M	173

Figure 6.6: Sample Height Data

Please note that the variable *enrolment number* need not be used in the analysis, so no summary data for the enrolment number will be performed.

Descriptive Analysis of Categorical Data:

Gender is a categorical variable in Figure 6. In this case, the summary would be the frequency table of various categories. For example, for the given data, the frequency distribution would be:

Gender	Frequency	Proportion	Percentage
Female (F)	5	0.5	50%
Male (M)	5	0.5	50%

In addition, you can draw a bar chart or pie chart to describe the data of the Gender variable. The pie chart for such data is shown in Figure 7. Details of different charts are explained in Unit 4. In general, you draw a bar graph if the number of categories exceeds three.

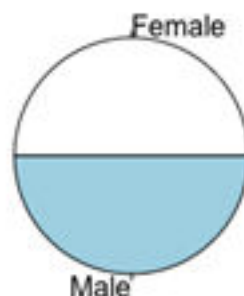


Figure 6.7: The Pie Chart

Descriptive analysis of Quantitative Data:

The height is a quantitative variable. The descriptive analysis of quantitative data is presented by the following:

1. Describing the central tendencies of the data
2. Describing the spread of the data.

Central tendencies of Quantitative data: Mean and Median are two basic measures that define the centre of data by using different ways. They are explained below with the help of an example.

Example 2: Find the mean and median of the following data:

Data Set (n observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	21	25	10	18	9	7	14	11	19	14

The mean can be computed as follows:

$$\bar{x} = \frac{\sum_{i=1}^n x}{n}$$

For the given data

$$\bar{x} = \frac{(4 + 21 + 25 + 10 + 18 + 9 + 7 + 14 + 11 + 19 + 14)}{11}$$

Mean, $\bar{x} = 13.82$

The median of the data would be the mid-value of the sorted data. First, data is sorted, and the median is computed using the following formula:

If n is even, then

$$\text{median} = [(Value\ of\ (\frac{n}{2})^{th}\ position) + Value\ of\ ((\frac{n}{2} + 1)^{th}\ position)]/2$$

If n is odd, then

$$\text{median} = (Value\ of\ (\frac{n+1}{2})^{th}\ position)$$

For this example, the sorted data is as follows:

Data Set (n observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	7	9	10	11	14	14	18	19	21	25

So, the median is:

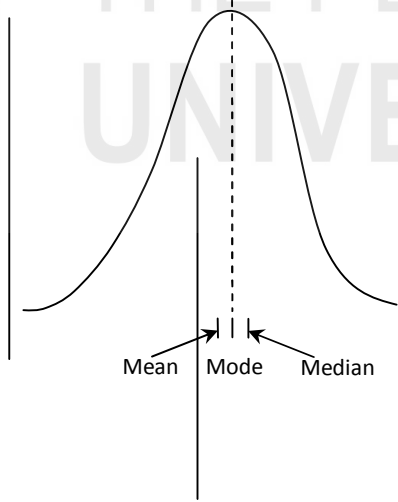
$$\text{median} = (\text{Value of } (\frac{11+1}{2})^{\text{th}}) \text{ position} = 14$$

Please note that outliers, defined as values highly different from most other values, can impact the mean but not the median. For example, if one observation in the data, as shown in example 2, is changed as:

Data Set (n observations)	1	2	3	4	5	6	7	8	9	10	11
x	4	7	9	10	11	14	14	18	19	21	100

Then, the median will remain at 14; however, the mean will change to 20.64, which is quite different from the earlier mean. Thus, you should be careful about the presence of outliers during data analysis.

Interestingly, mean and mode may be useful in determining the nature of data. The following table describes these conditions:

Relationship between mean and mode	Comments about observations	A Possible Graph of Data Distribution
Almost Equal values of mean and median	The distribution of data may be symmetric	

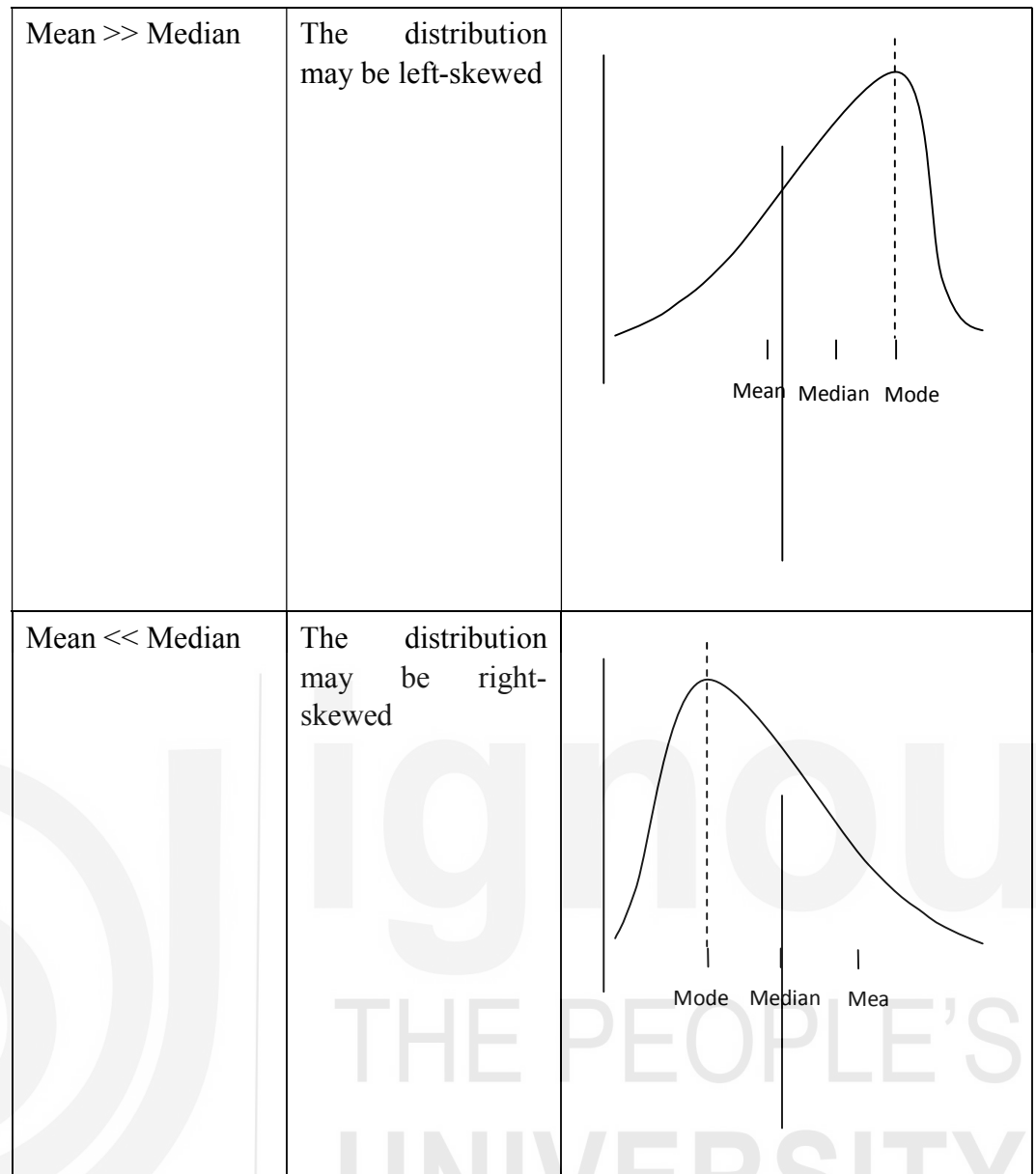


Figure 6.8: Mean and Median for possible data

The concept of data distribution is explained in the next Unit.

Mode: Mode is defined as the most frequent value of a set of observations. For example, in the data of example 2, the value 14, which occurs twice, is the mode. The mode value need not be a mid-value; rather, it can be any value of the observations. It just communicates the most frequently occurring value only. In a frequency graph, the mode is represented by the peak of the data. For example, in the charts shown in Figure 8, the value corresponding to the peaks is the mode.

Spread of Quantitative Data: Another important aspect of defining quantitative data is its spread or variability of observed data. Some of the measures for the spread of data are given in Figure 9.

Measure	Description	Example (Please refer to Data of Example 2)
Range	Minimum to Maximum Value	4 to 25
Variance	The sum of the squares of the difference between the observations and its sample mean is divided by $(n-1)$, as the difference of n^{th} value can be determined from $(n-1)$ computed differences since the overall sum of differences has to be zero. The formula of variance for a sample is: $s^2 = \frac{1}{(n-1)} \sum_{i=1}^n (x - \bar{x})^2$ However, in case you are determining the Population variance, then you can use the following formula: $\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x - \mu)^2$	Try both the formula and then match the answer: 40.96
Standard Deviation	Standard deviation is one of the most used measures for finding the spread or variability of data. It can be computed as: For Sample: $s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x - \bar{x})^2}$ For Population: $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n (x - \mu)^2}$	Try both the formula and then match the answer: 6.4
5-Point Summary and Interquartile Range (IQR)	To create a 5-point summary, first, you need to sort the data. The five-point summary is defined as follows: Minimum Value (Min) 1st Quartile $\leq$ 25% values (Q1) 2nd Quartile is median (M) 3rd Quartile is $\leq$ 75% values (Q3) Maximum Value (Max) IQR is the difference between the values of the 3rd and 1st quartiles.	Use Sorted data from Example 2 Min = 4 Q1 = (9+10)/2 = 9.5 M = 14 Q3 = (18+19)/2 = 18.5 Max = 25 IQR = 18.5 - 9.5 = 9

Figure 6.9: The Measure of Spread or Variability

The IQR can also be used to identify suspected outliers. In general, a suspected outlier can exist in the following two ranges:

Observation/values less than $Q1 - 1.5 \cdot IQR$

Observation/values more than $Q3 + 1.5 \cdot IQR$

For example 2, IQR is 9; therefore, the outliers may be: Values $< (9.5 - 9)$ or Values < 0.5 . or Values $> (18.5 - 9)$ or Values > 27.5 .

Thus, there is no outlier in the initial data of Example 2.

You may draw various plots for the qualitative data, such as histogram, boxplot, etc. These plots are explained in Unit 4 of this block.

Check Your Progress 2

1. The age category of students is categorical data. What information would you like to show for its descriptive analysis?

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2. Age is quantitative data; how will you describe its data?

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3. How can you find that given data is left skewed?

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4. What is IQR? Can it be used to find outliers?

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6.4.2 Exploratory Analysis

Exploratory data analysis was suggested by John Turkey of Princeton University in 1960 as a group of methods that can be used to learn the possibilities of relationships amongst data. After obtaining relevant data for analysis, instead of performing the final analysis, you may like to explore the data for possible relationships using exploratory data analysis. In general, graphs are some of the best ways to perform exploratory analysis. Some of the standard methods that you can perform during exploratory analysis are as follows:

1. As a first step, you may perform the descriptive analysis of various categorical and qualitative variables of your data. Such information is

very useful in determining the suitability of data for analysis. This may also help you in data cleaning, modification and transformation of data.

- a. For the qualitative data, you may create frequency tables and bar charts to know the distribution of data among different categories. A balanced distribution of data among classes is most desirable. However, such distribution may not be possible in actual situations. Several methods have been suggested to deal with such situations. Some of those will be discussed in the later units.
 - b. For the quantitative data, you may compute the mean, median, standard deviation, skewness and kurtosis. The kurtosis value relates to peaks (determined by mode) in the data. In addition, you may also draw the charts like histogram to look into frequency distribution.
2. Next, after performing the univariate analysis, you may try to perform some bi-variate analysis. Some of the basic statistics that you can perform for bi-variate analysis include the following:
- a. Make a two-way table between categorical variables and make related stacked bar charts. You may also use chi-square testing to find any significant relationships.
 - b. You may draw side-by-side box plots to check if the data of various categories have differences.
 - c. You may draw a scatterplot and check the correlation coefficient if that exists between two variables.
3. Finally, you may like to look into the possibilities of multi-variate relationships among data. You may use dimensionality reduction by using techniques like feature extraction or principle component analysis, you may perform clustering to identify possible sets of classes in the solution space, or use graphical tools, like bubble charts, to visualise the data.

It may be noted that exploratory data analysis helps in identifying the possibilities of relationships amongst data but does not promise that a causal relationship may exist amongst variables. The causal relationship has to be ascertained through qualitative analysis. Let us explain the exploratory data analysis with the help of an example.

Example 3: Consider the sample data of students given in Figure 6 about Gender and Height. Let us explore this data for an analytical question: Does height depend on Gender?

You can perform the exploratory analysis on this data by drawing a side-by-side boxplot for Male and Female students' height. This boxplot is shown in Figure 6.10.

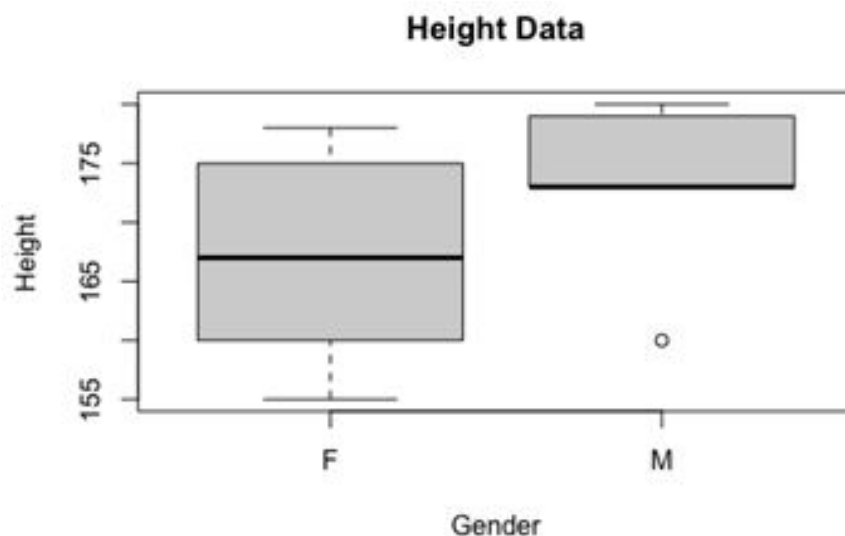


Figure 6.10: Exploratory Data Analysis

Please note that the box plot of Figure 10 shows that the average height of male students is more than that of female students. Does this result apply, in general, to the population? To answer this question, you need to find the probability of the occurrence of such sample data, thereafter, you may perform an Inferential analysis.

Check Your Progress 3

1. What is the advantage of using a boxplot?

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6.5 SUMMARY

This Unit introduces basic statistical and analytical concepts of data science. This Unit first introduces you to the definition of data science. Data science uses concepts from computing, mathematics and domain knowledge. The types of data for data science are defined in two different ways. First, it is defined based on the structure and generation rate of data. Next, it is defined as the measures that can be used to capture the data. In addition, the concept of sampling has been defined in this Unit.

This Unit also explains some of the basic analysis techniques, including descriptive, exploratory, inferential and predictive. A few misconceptions related to data science have also been explained with the help of examples. This Unit also introduces you to some of the applications of data science and the data science life cycle. In the ever-advancing technology, it is suggested to keep reading about newer data science applications.

Check Your Progress 1:

1. Data science integrates the principles of computer science, mathematics and domain knowledge to create mathematical models that show relationships among data attributes. In addition, data science uses data to perform predictive analysis.
2. Structured data has a defined dimensional structure identified by attributes, for example, tables, data cubes, etc. Semi-structured data has some structure due to the use of tags; however, this structure may be flexible, for example, XML data. Unstructured data has no structure at all, like long texts. Data streams, on the other hand, may be structured, semi-structured or unstructured data being produced continuously.
3. Age category would be categorical data, it will be of the ordinal scale, as there are differences among categories, but that difference cannot be defined quantitatively. The weight of the students in a class is a ratio scale. The Grade is also a measure of the ordinal scale. The 5-point Likert scale is also ordinal data.

Check Your Progress 2:

1. Descriptive analysis of categorical data may include the total number of observations, frequency table and bar or pie chart.
2. The descriptive age may include mean, median, mode, skewness, kurtosis, standard deviation and histogram or box plot.
3. For left-skewed data, the mean is substantially higher than the median and mode.
4. The difference between the Quartile 3 and Quartile 1 is the interquartile range (IQR). In general, suspected outliers are at a distance of 1.5 times IQR higher than the 3rd quartile or 1.5 times IQR lower than the 1st quartile.

Check Your Progress 3:

1. Boxplot shows a 5-point summary of data. A well-spread box plot is an indicator of normally distributed data. Side-by-side boxplots can be used to make a comparison of scale data values of two or more categories.